

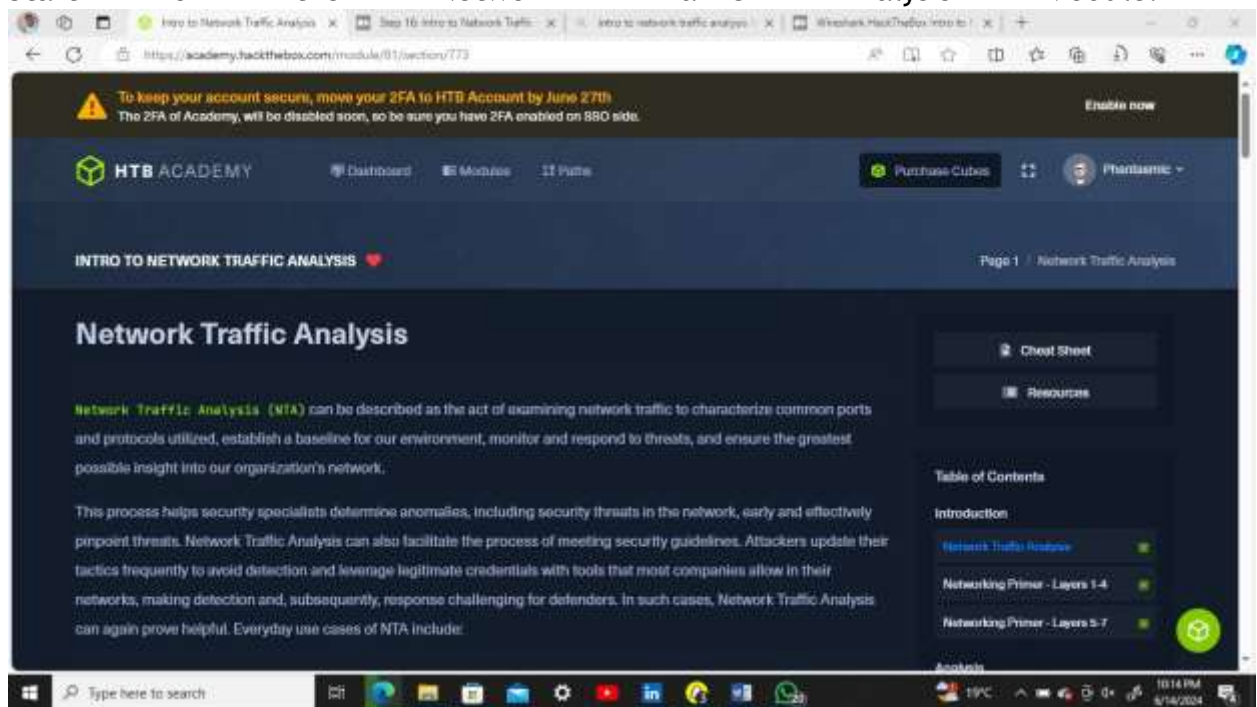
Hack The Box Intro to Network Traffic Analysis Module

Report by Ezekiel Waweru CNS06-24049

Introduction

This hackthebox module offered a better understanding into how to capture, analyze and filter network packets using tools like Wireshark, Tshark, and TCP dump. Step by step, I got to understand how to use them to capture network traffic for different protocols, analyze it by reading the logs, and make a decision if the traffic is malicious

I. Start of the Network Traffic Analysis Module.



II. Networking Primer Layer 1-4 questions

This module looked at basic network flows through the 4 layers and the protocols used. The OSI and TCP/IP models were examined and contrasted giving me a view on how each one of them functions. Protocol Data Units were examined in depth. Addressing and transport mechanisms sections offered more insight into critical networking fundamentals.

Intro to Network Traffic Analysis

https://academy.hackthebox.com/module/81/section/954

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Answer the questions below to complete this section and earn cups!

0 How many layers does the OSI model have?

7

Submit Hint

0 How many layers are there in the TCP/IP model?

4

Submit Hint

0 True or False: Routers operate at layer 2 of the OSI model?

False

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12:38 AM 6/15/2024

Intro to Network Traffic Analysis

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0 What addressing mechanism is used at the Link Layer of the TCP/IP model?

MAC address

Submit Hint

0 At what layer of the OSI model is a PDU encapsulated into a packet? (the number)

3

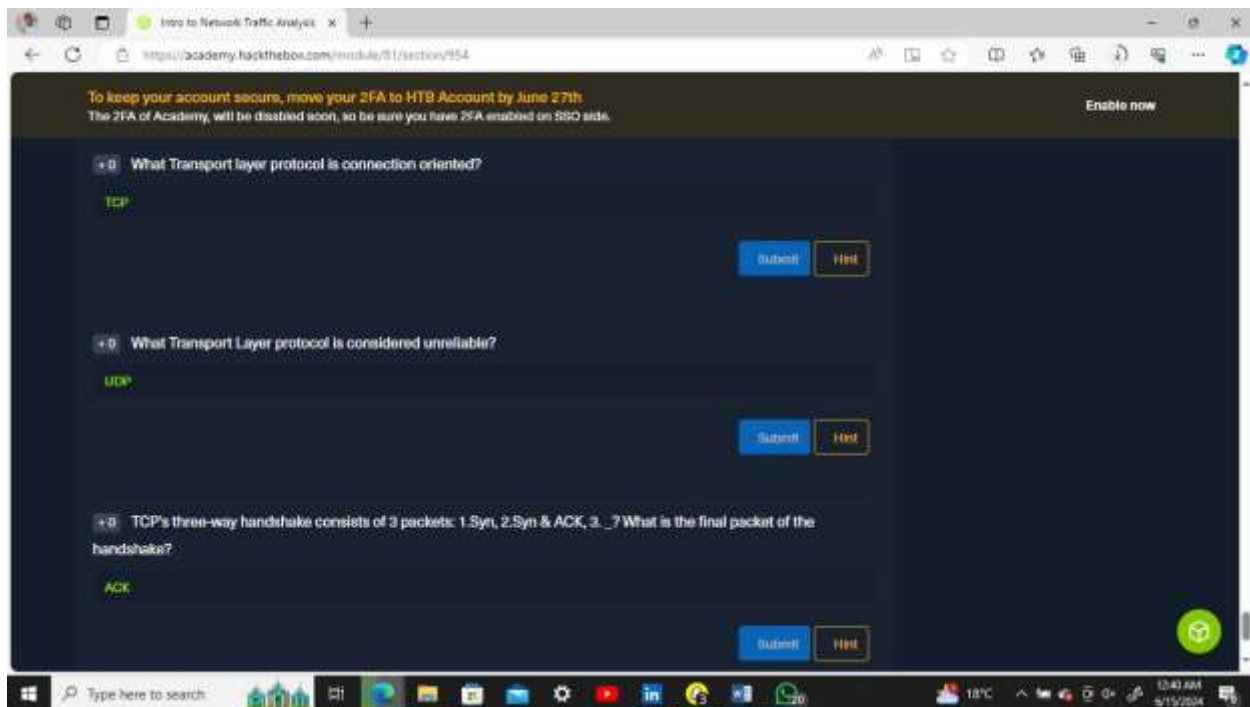
Submit Hint

0 What addressing mechanism utilizes a 32-bit address?

IPv4

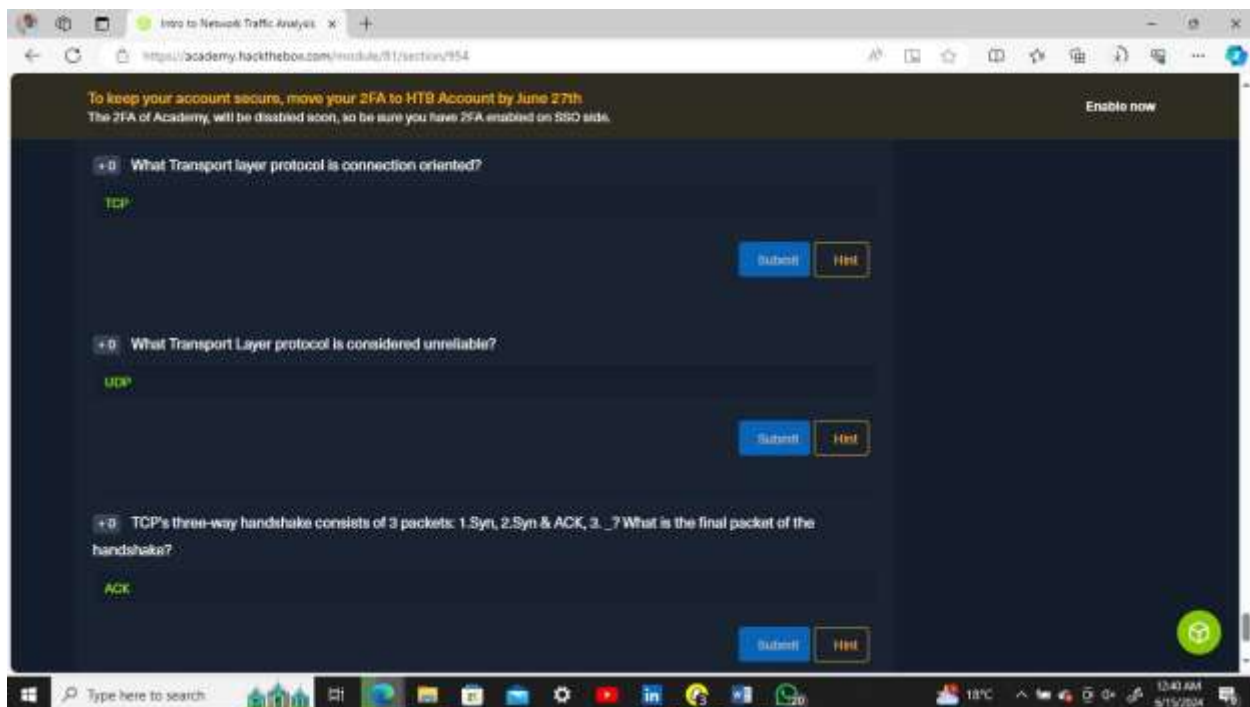
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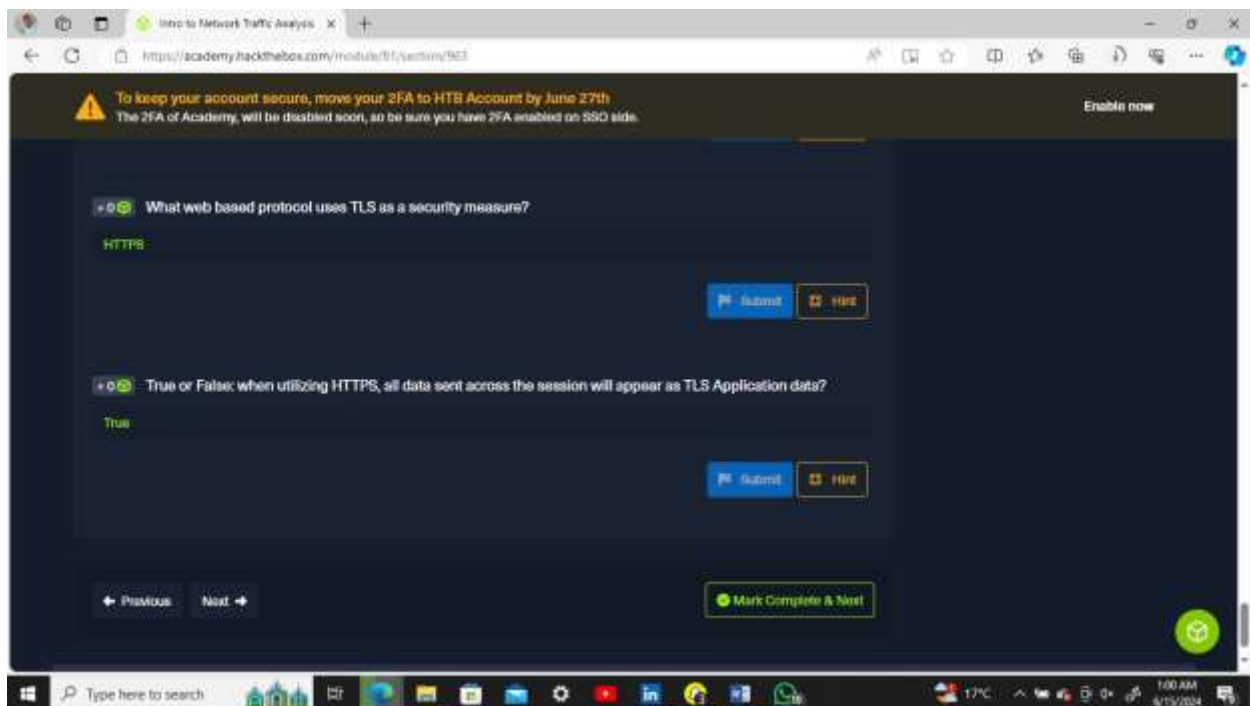
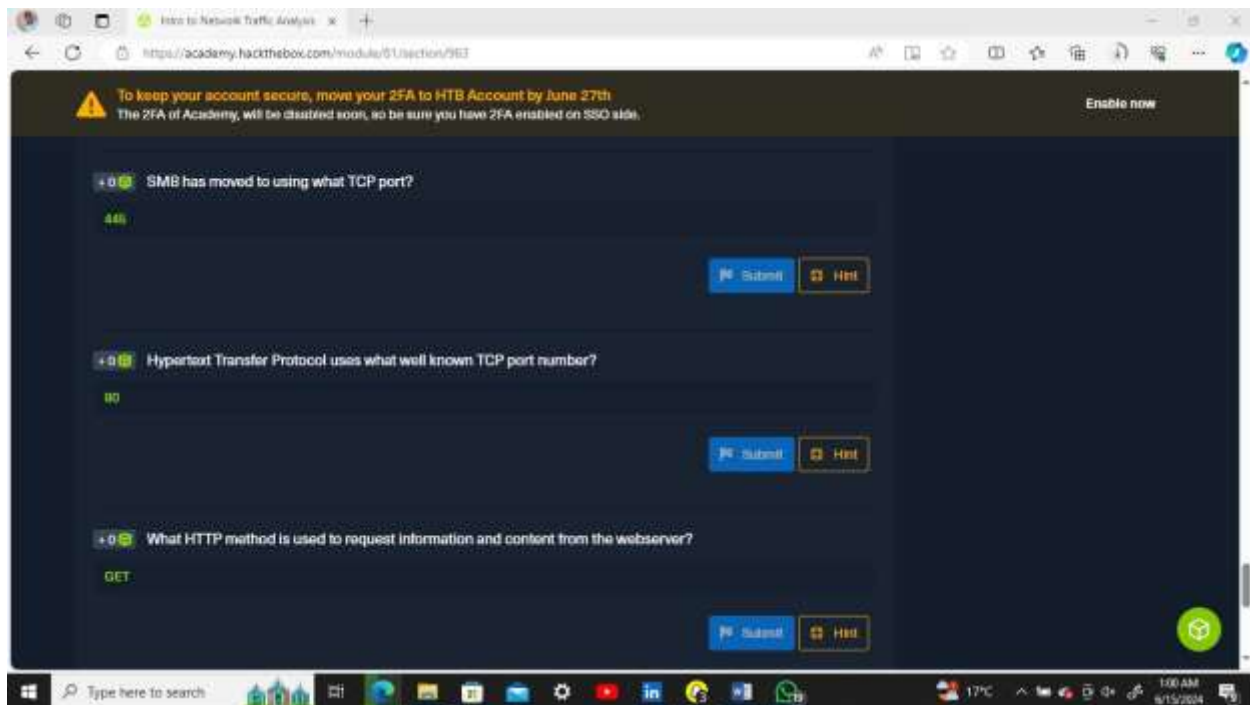
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III. Networking Primer Layer 5-7

Got to learn about higher level networking protocols





IV. TCP Dump Fundamentals.

https://academy.hackthebox.com/module/81/function/774

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```
File Edit View Search Terminal Help
[eu-academy-1]-[10.10.14.172]-[htb-ac-1074699@htb-ya5m8r3gxl]-[~]
[*]$ sudo tcpdump --version
tcpdump version 4.99.3
libpcap version 1.10.3 (with TPACKET_V3)
n0penSSL 1.1.1n 15 Mar 2022
[eu-academy-1]-[10.10.14.172]-[htb-ac-1074699@htb-ya5m8r3gxl]-[~]
[*]$ man tcpdump
[eu-academy-1]-[10.10.14.172]-[htb-ac-1074699@htb-ya5m8r3gxl]-[~]
[*]$ sudo tcpdump -D
1.eth0 [Up, Running, Connected]
2.tun0 [Up, Running, Connected]
3.any (Pseudo-device that captures on all interfaces) [Up, Running]
4.lo [Up, Running, Loopback]
5.bluetooth-monitor (Bluetooth Linux Monitor) [Wireless]
6.nflog (Linux netfilter log (NFLOG) interface) [none]
7.nfqueue (Linux netfilter queue (NFQUEUE) interface) [none]
8.dbus-system (D-Bus system bus) [none]
9.dbus-session (D-Bus session bus) [none]
[eu-academy-1]-[10.10.14.172]-[htb-ac-1074699@htb-ya5m8r3gxl]-[~]
[*]$ sudo tcpdump -i eth0
```

Integrated Terminal

https://academy.hackthebox.com/module/81/function/774

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```
Applications Places System
Parrot Terminal
File Edit View Search Terminal Help
), ethertype IPv4 (0x0800), length 66: 94-237-60-223.uk-htb1.upcloud.host.39646
> htb-ya5m8r3gxl.htb-cloud.com.http: Flags [P.], ack 887953, win 9070, options [n
op,nop,T5 val 2733639087 ecr 1909835138], length 0
00:17:39.609551 a6:ba:3b:08:6d:04 (oui Unknown) > 9a:7e:00:3f:c0:b8 (oui Unknown
), ethertype IPv4 (0x0800), length 41071: htb-ya5m8r3gxl.htb-cloud.com.http > 94
n.237-60-223.uk-htb1.upcloud.host.39646: Flags [P.], seq 887953:928958, ack 161,
win 501, options [nop,nop,T5 val 1909835138 ecr 2733639087], length 41005: HTTP
00:17:39.609854 9a:7e:00:3f:c0:b8 (oui Unknown) > a6:ba:3b:08:6d:04 (oui Unknown
), ethertype IPv4 (0x0800), length 66: 94-237-60-223.uk-htb1.upcloud.host.39646
> htb-ya5m8r3gxl.htb-cloud.com.http: Flags [P.], ack 898089, win 9300, options [n
op,nop,T5 val 2733639088 ecr 1909835138], length 0
00:17:39.609855 9a:7e:00:3f:c0:b8 (oui Unknown) > a6:ba:3b:08:6d:04 (oui Unknown
), ethertype IPv4 (0x0800), length 66: 94-237-60-223.uk-htb1.upcloud.host.39646
> htb-ya5m8r3gxl.htb-cloud.com.http: Flags [P.], ack 928958, win 9172, options [n
op,nop,T5 val 2733639088 ecr 1909835138], length 0
^C
301 packets captured
304 packets received by filter
0 packets dropped by kernel
[eu-academy-1]-[10.10.14.172]-[htb-ac-1074699@htb-ya5m8r3gxl]-[~]
[*]$
```

Integrated Terminal

🔒 Hack The Box - Academy

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🔒 Utilizing the output shown in question-1.png, who is the server in this communication? (IP Address)

174.143.213.194

[Submit](#) [question-1.zip](#) [Hint](#)

🔒 Were absolute or relative sequence numbers used during the capture? (see question-1.zip to answer)

relative

[Submit](#) [question-1.zip](#) [Hint](#)

🔒 If I wish to start a capture without hostname resolution, verbose output, showing contents in ASCII and hex, and grab the first 100 packets; what are the switches used? please answer in the order the switches are asked for in the question.

-nXc 100

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🔒 Given the capture file at /tmp/capture.pcap, what tcpdump command will enable you to read from the capture and show the output contents in Hex and ASCII? (Please use best practices when using switches)

sudo tcpdump -R /tmp/capture.pcap

[Submit](#) [Hint](#)

🔒 What TCPDump switch will increase the verbosity of our output? (Include the - with the proper switch)

-v

[Submit](#) [Hint](#)

🔒 What built in terminal help reference can tell us more about TCPDump?

man

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👤 Given the capture file at /tmp/capture.pcap, what tcpdump command will enable you to read from the capture and show the output contents in Hex and ASCII? (Please use best practices when using switches)

```
sudo tcpdump -r /tmp/capture.pcap
```

[Submit](#) [Hint](#)

👤 What TCPDump switch will increase the verbosity of our output? (Include the - with the proper switch)

+

[Submit](#) [Hint](#)

👤 What built in terminal help reference can tell us more about TCPDump?

```
man
```

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man

[Submit](#) [Hint](#)

👤 What TCPDump switch will let me write my output to a file?

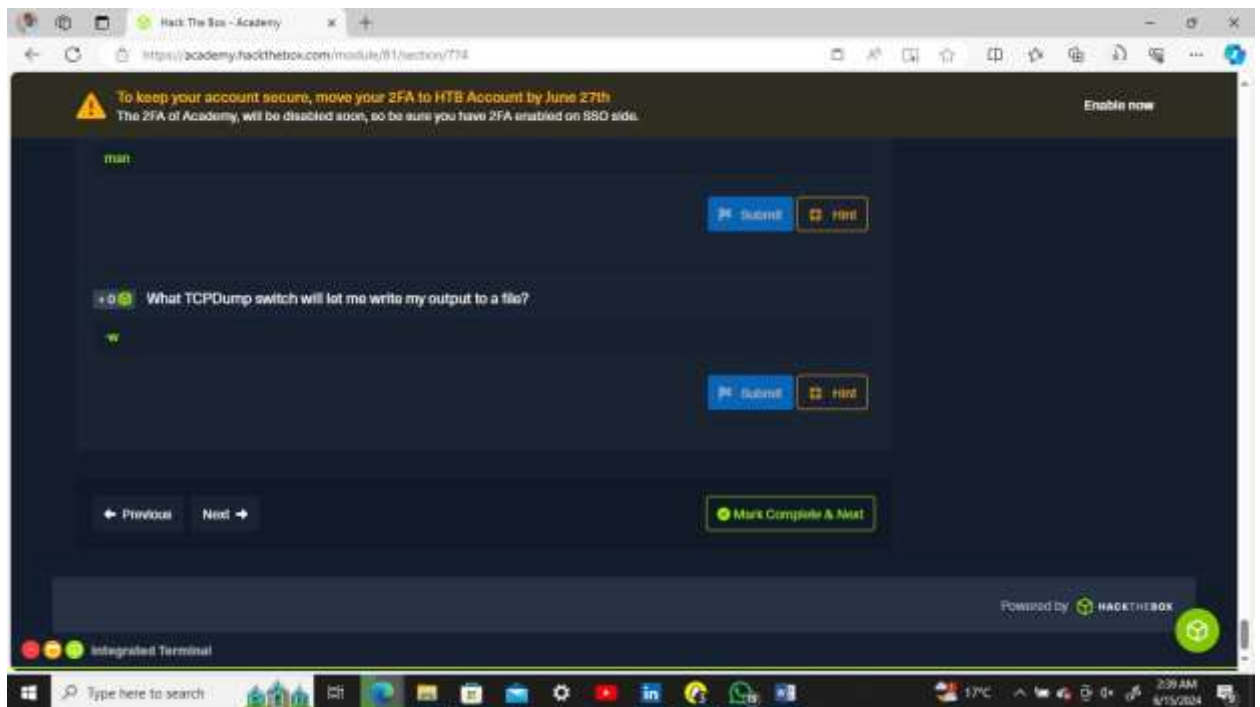
```
w
```

[Submit](#) [Hint](#)

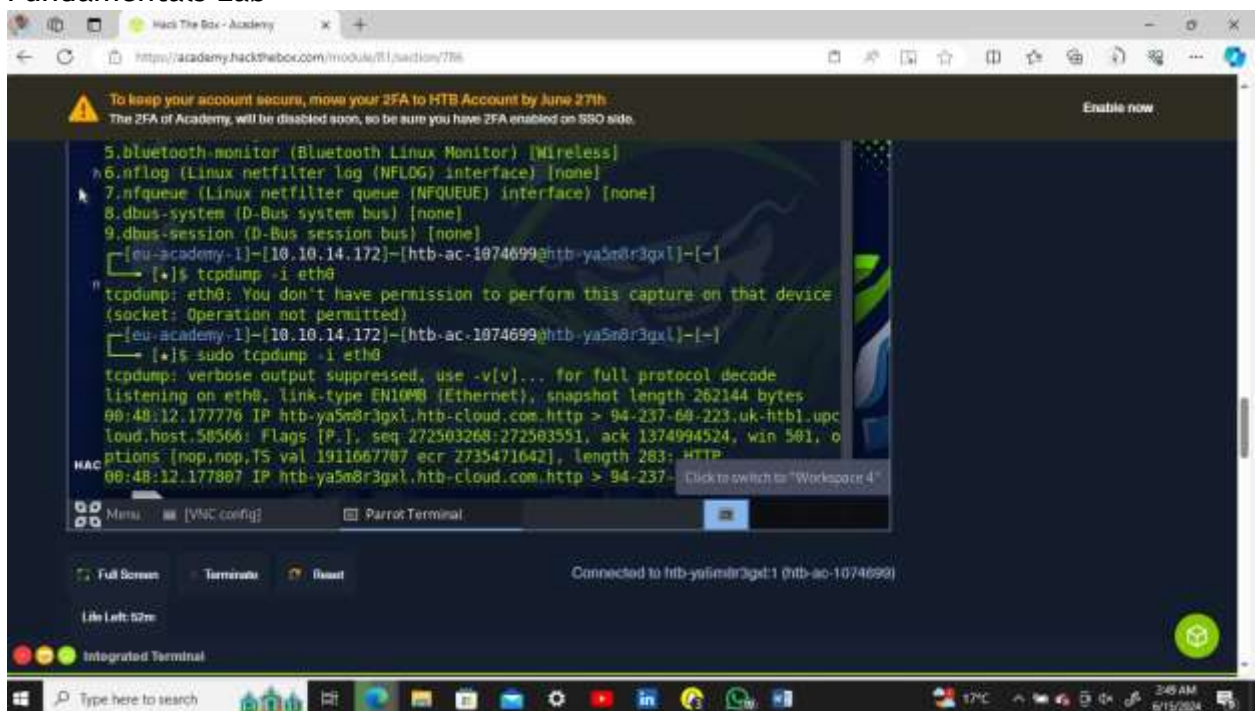
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V. Fundamentals Lab



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🔒 What TCPDump switch will allow us to pipe the contents of a pcap file out to another function such as 'grep'?

`-i`

[Submit](#) [Hint](#)

🔒 True or False: The filter 'port' looks at source and destination traffic.

`true`

[Submit](#) [Hint](#)

🔒 If we wished to filter out ICMP traffic from our capture, what filter could we use? (word only, not symbol please.)

`not icmp`

[Submit](#) [Hint](#)

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🔒 What command will show you where / if TCPDump is installed?

`which tcpdump`

[Submit](#) [Hint](#)

🔒 How do you start a capture with TCPDump to capture on eth0?

`tcpdump -i eth0`

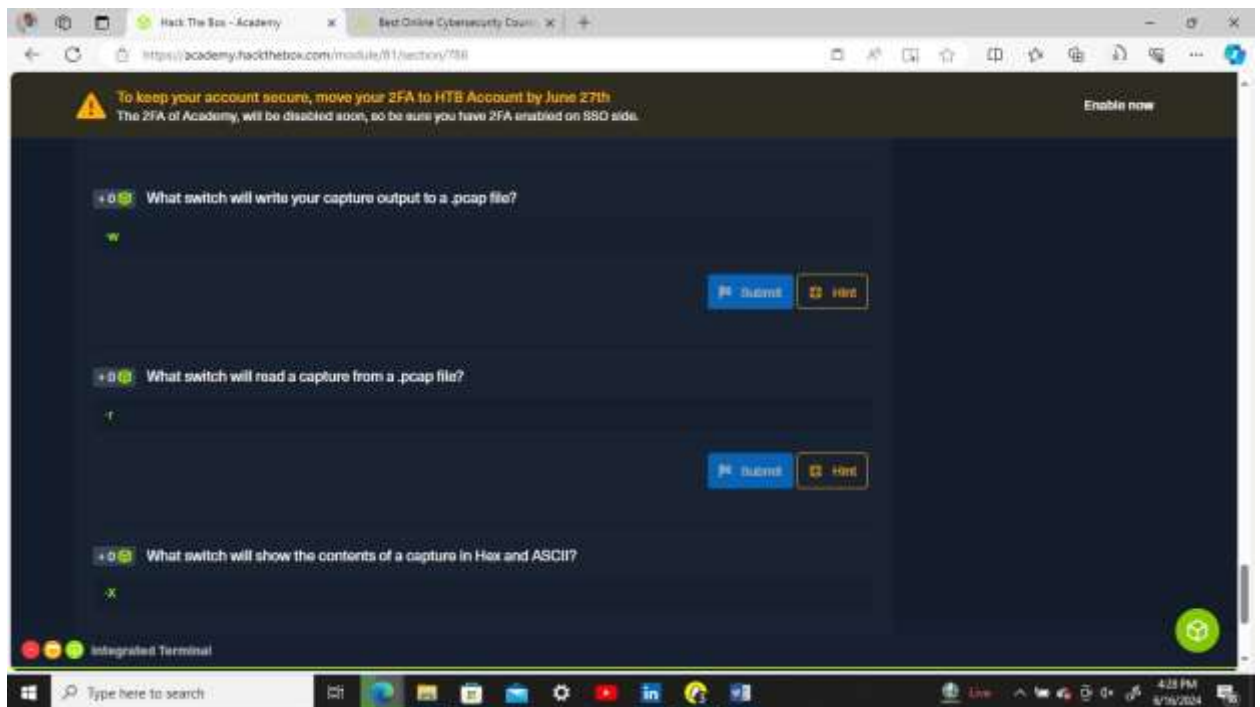
[Submit](#) [Hint](#)

🔒 What switch will provide more verbosity in your output?

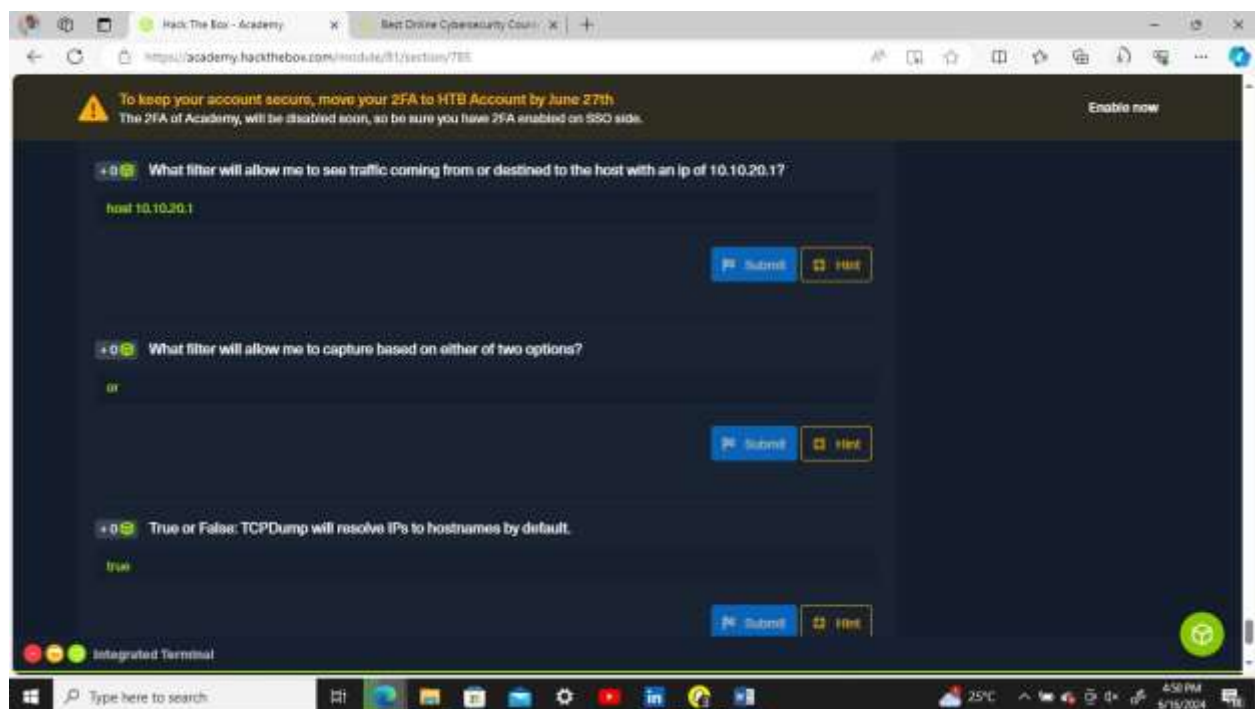
`-v`

[Submit](#) [Hint](#)

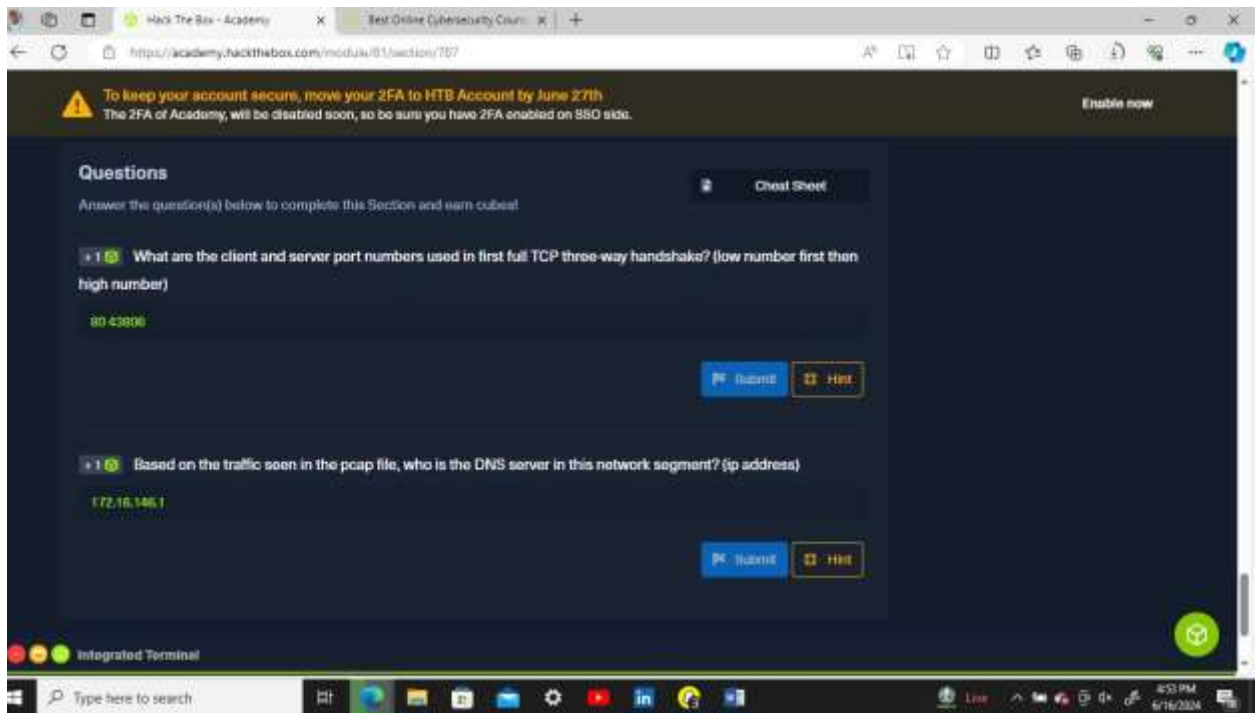
🔒 Integrated Terminal



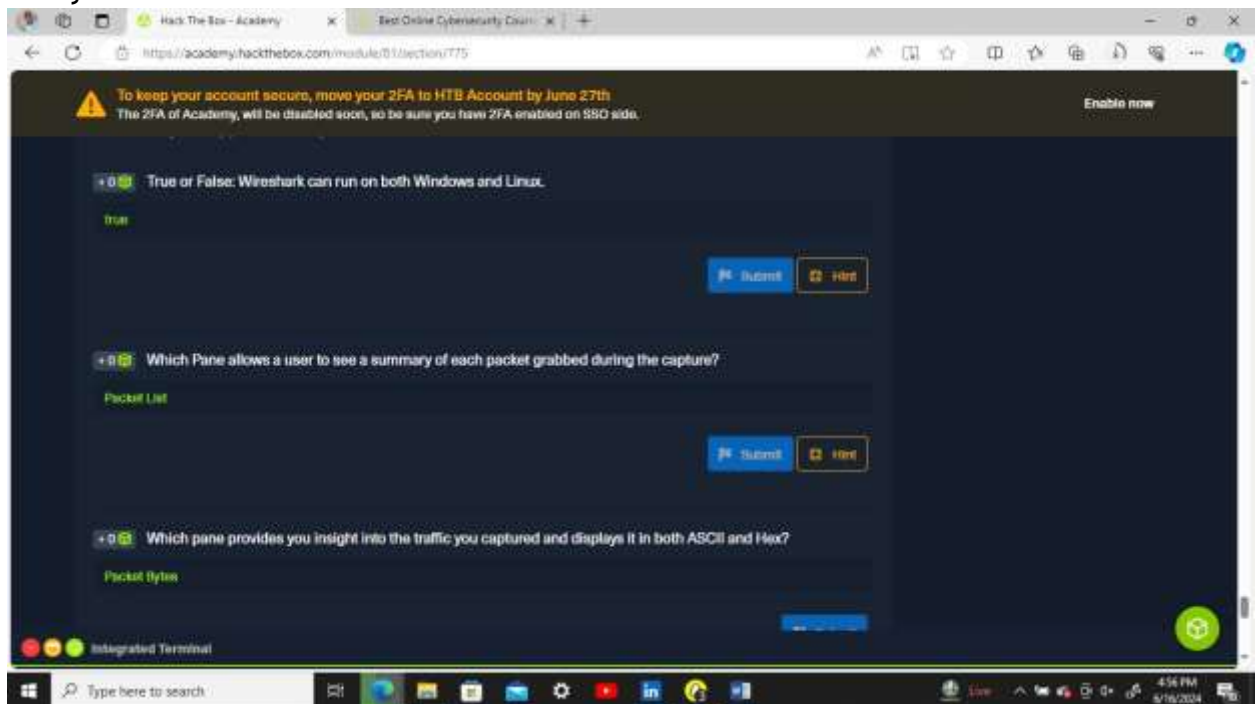
VI. TCPdump packet filtering

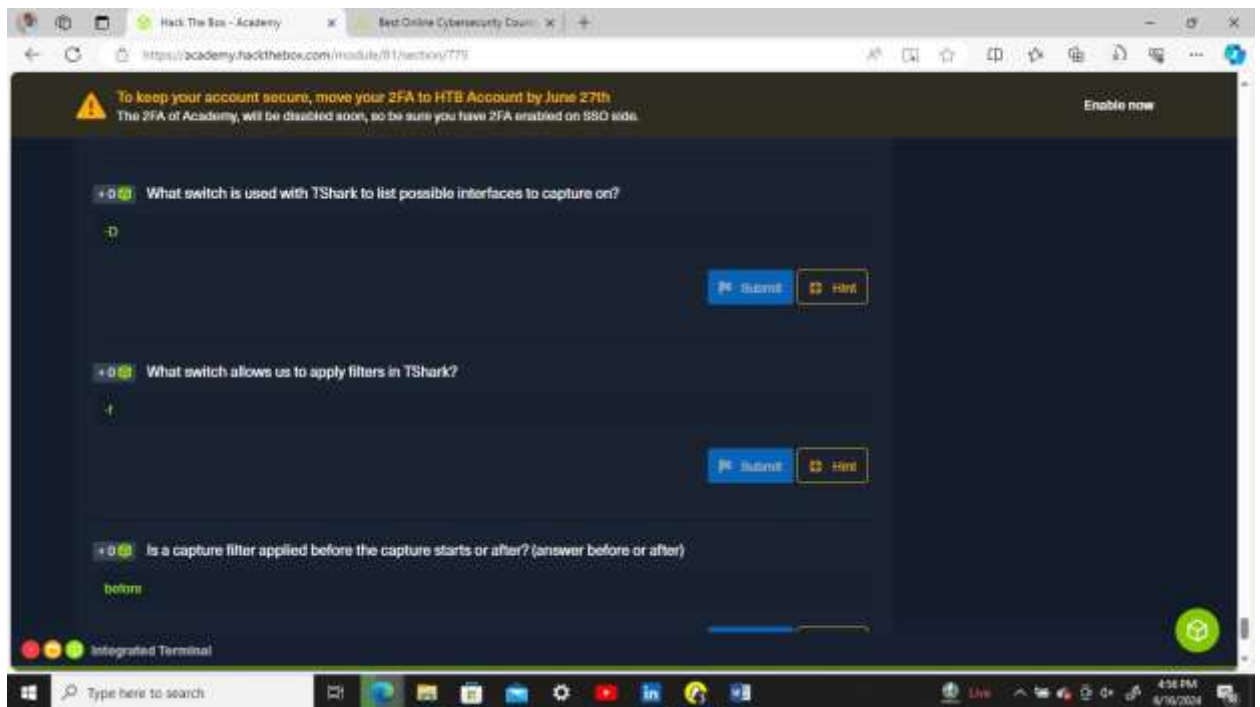


VII. Integrating network traffic with capture and delay filters.

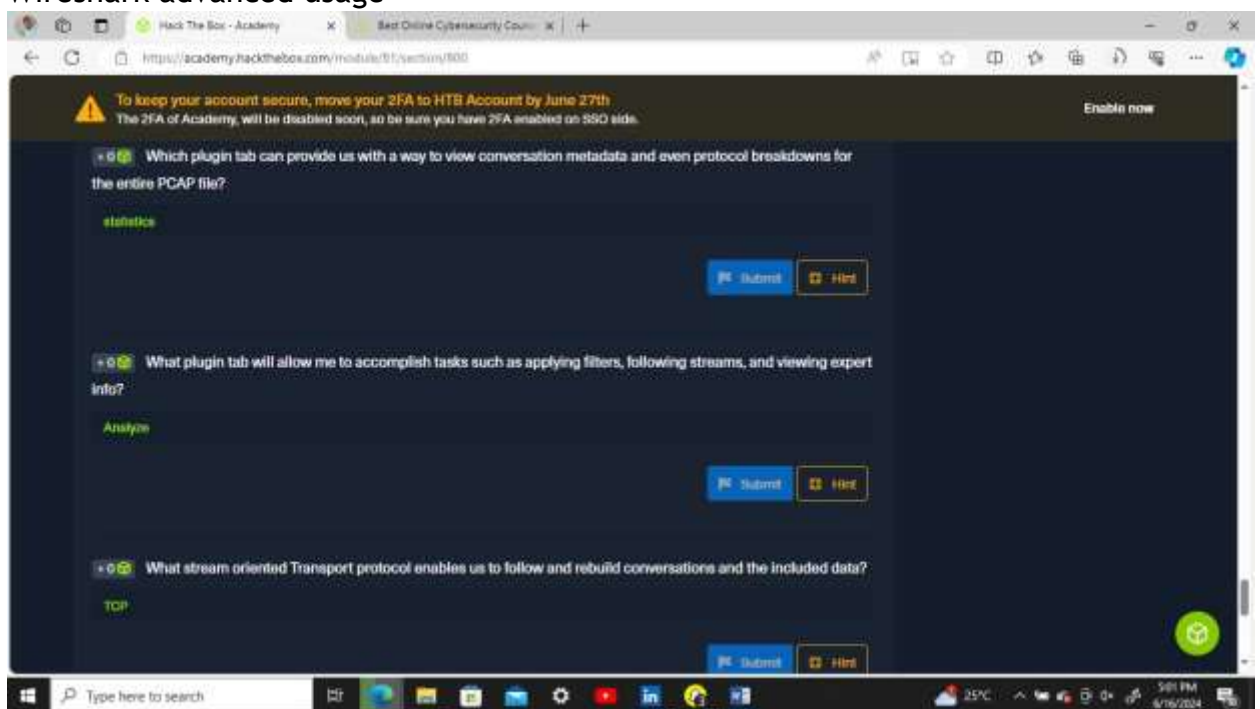


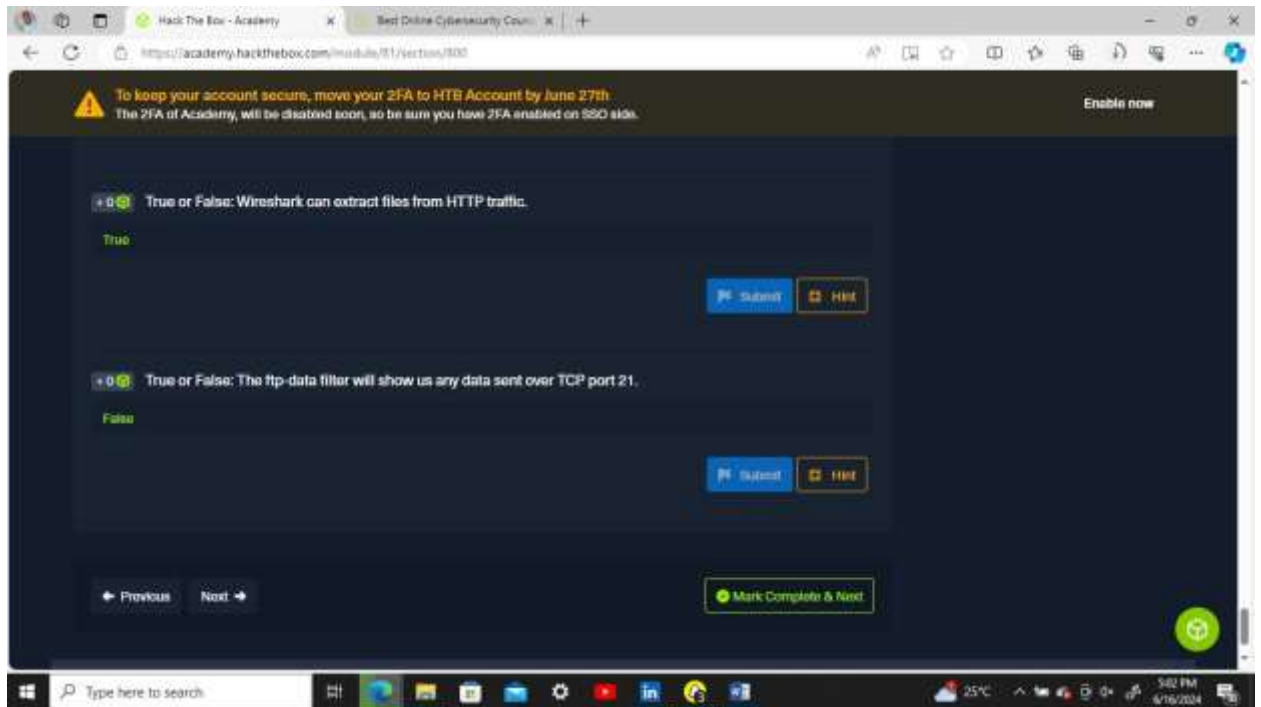
VIII. Analysis with Wireshark



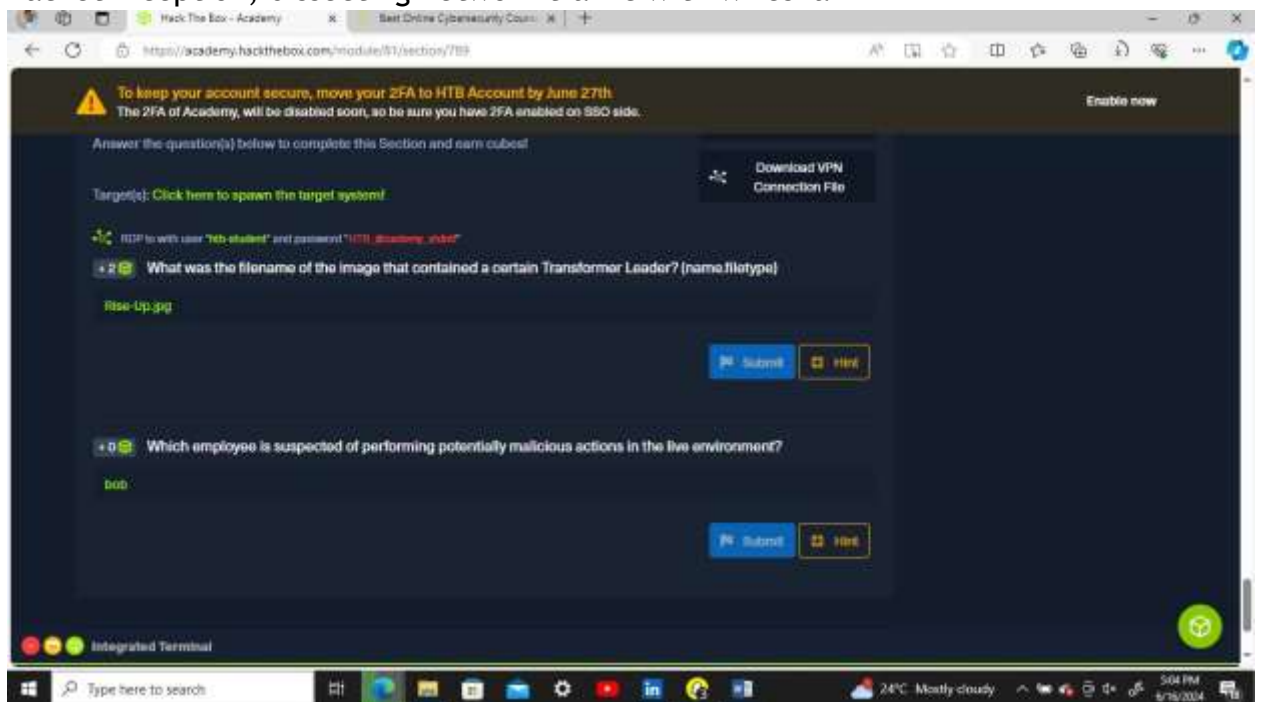


IX. Wireshark advanced usage

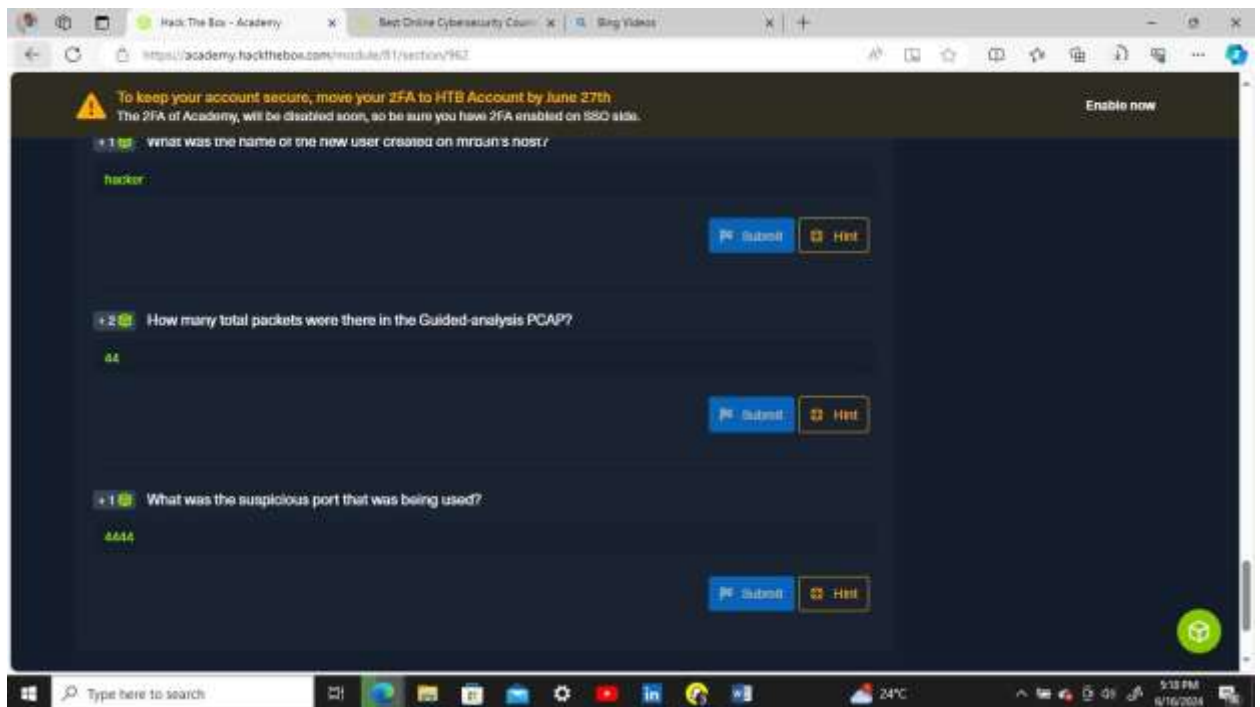




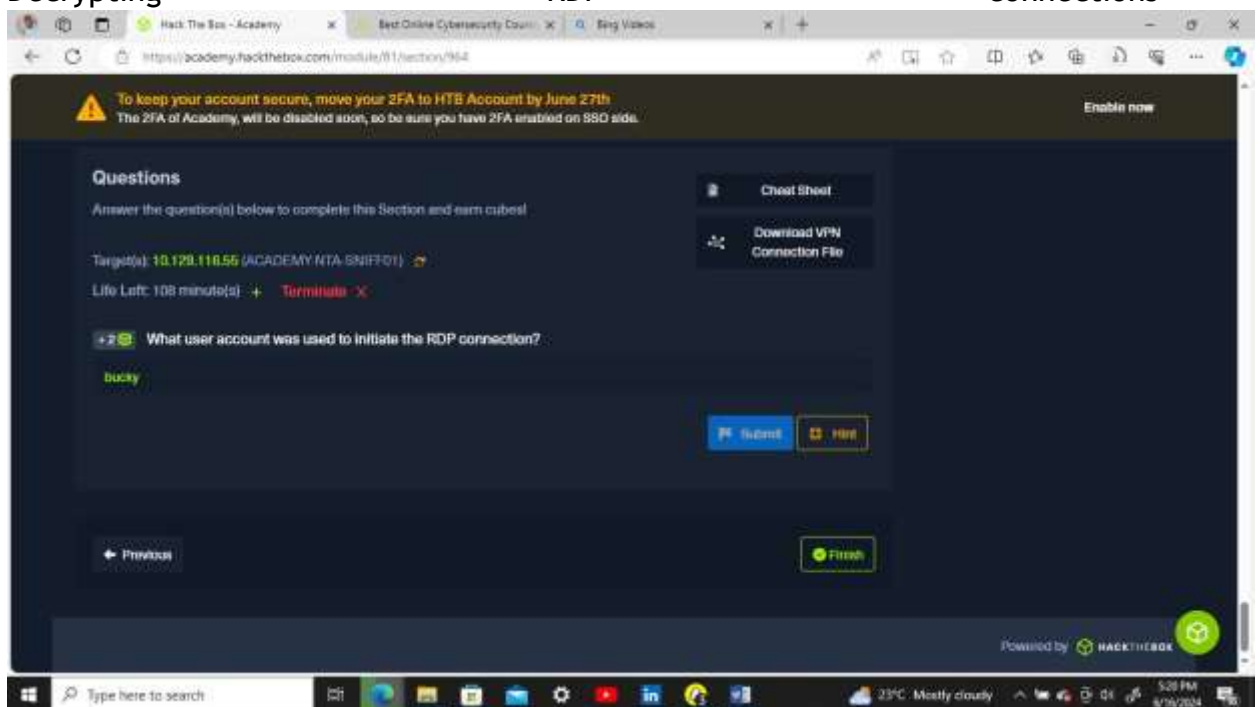
X. Packet inception, dissecting network traffic with wireshark



XI. Traffic analysis flow



XII. Decrypting RDP connections



Conclusion

Having completed this assignment, I have by the knowledge acquired here through hands on labs and material offered, got to know communication and traffic flow occurs

in the different TCP/IP and OSI layers. Getting to know that and how to examine packets through tools like Wireshark and TCPdump, has made me better equipped to secure a network against any malicious attacks.

Shareable link: <https://academy.hackthebox.com/achievement/1074699/81>